

# ■ Hospital OPD Analyzer

Weekly Performance & Overcrowding Report

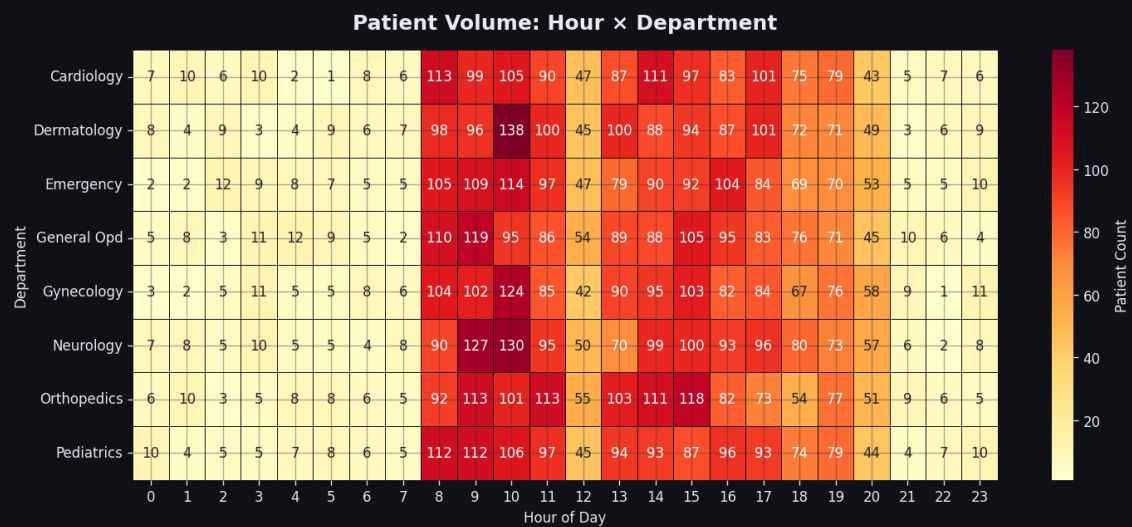
Data Period: 2024-01-01 00:00:00 → 2024-03-30 00:00:00

Generated: 21 April 2026, 03:04 PM

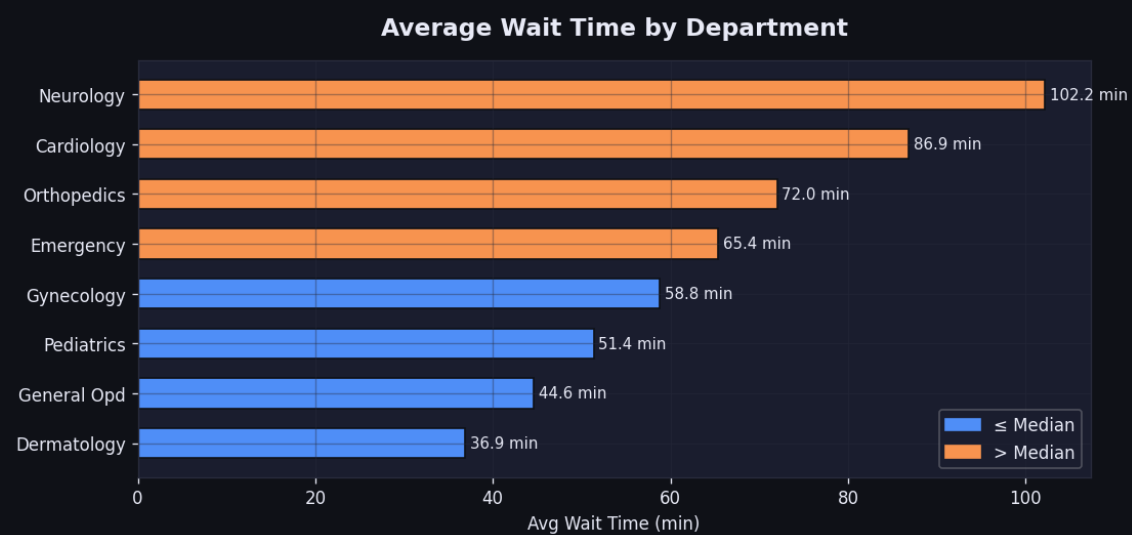
| Total Records | Departments | Date Range                                |
|---------------|-------------|-------------------------------------------|
| 9602          | 8           | 2024-01-01 00:00:00 – 2024-03-30 00:00:00 |

# Visualizations

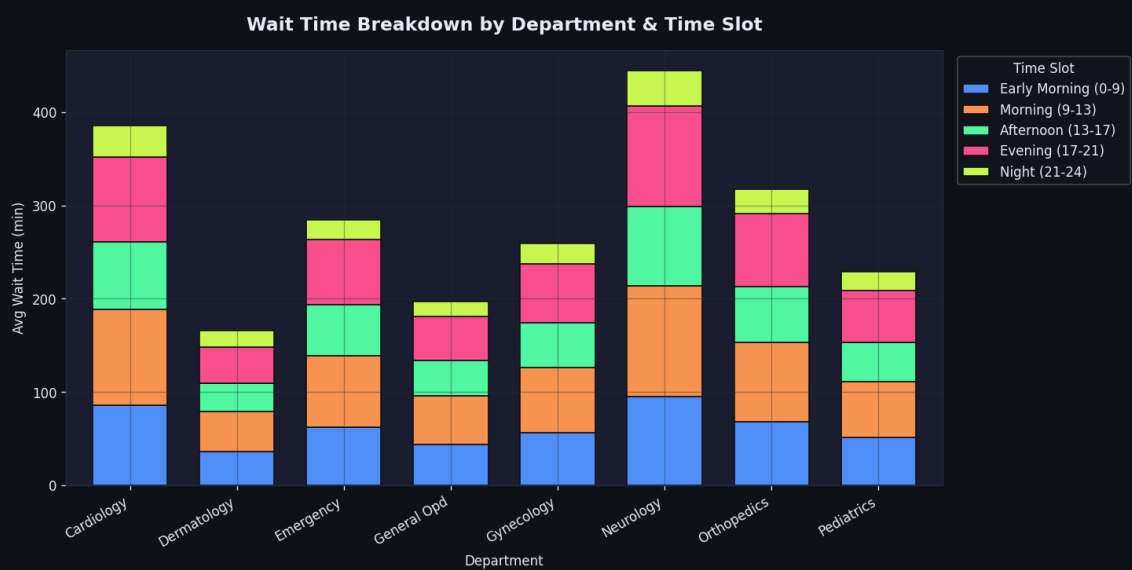
Patient Volume Heatmap (Hour × Department)



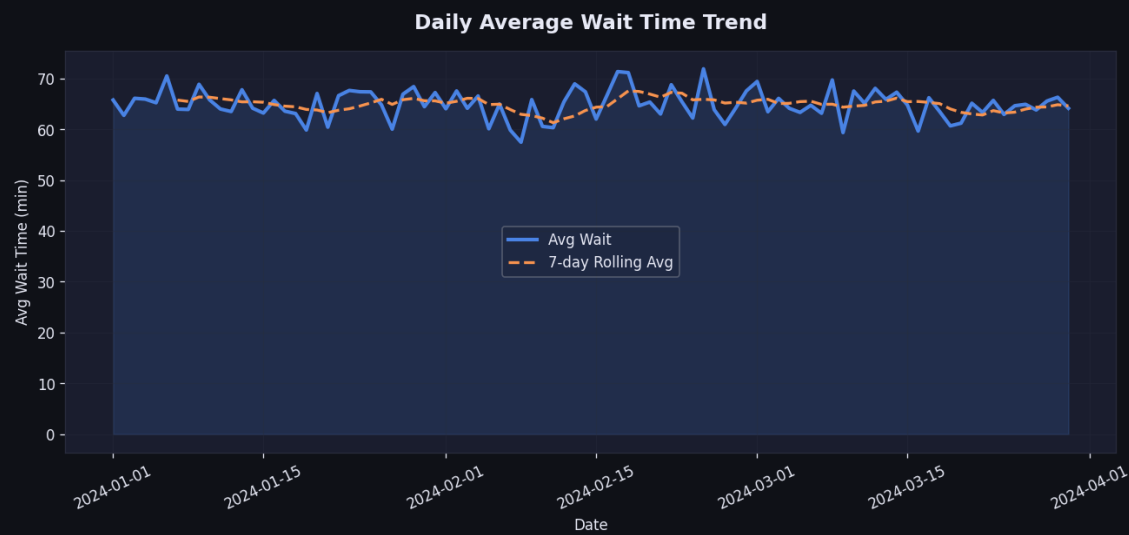
Average Wait Time by Department



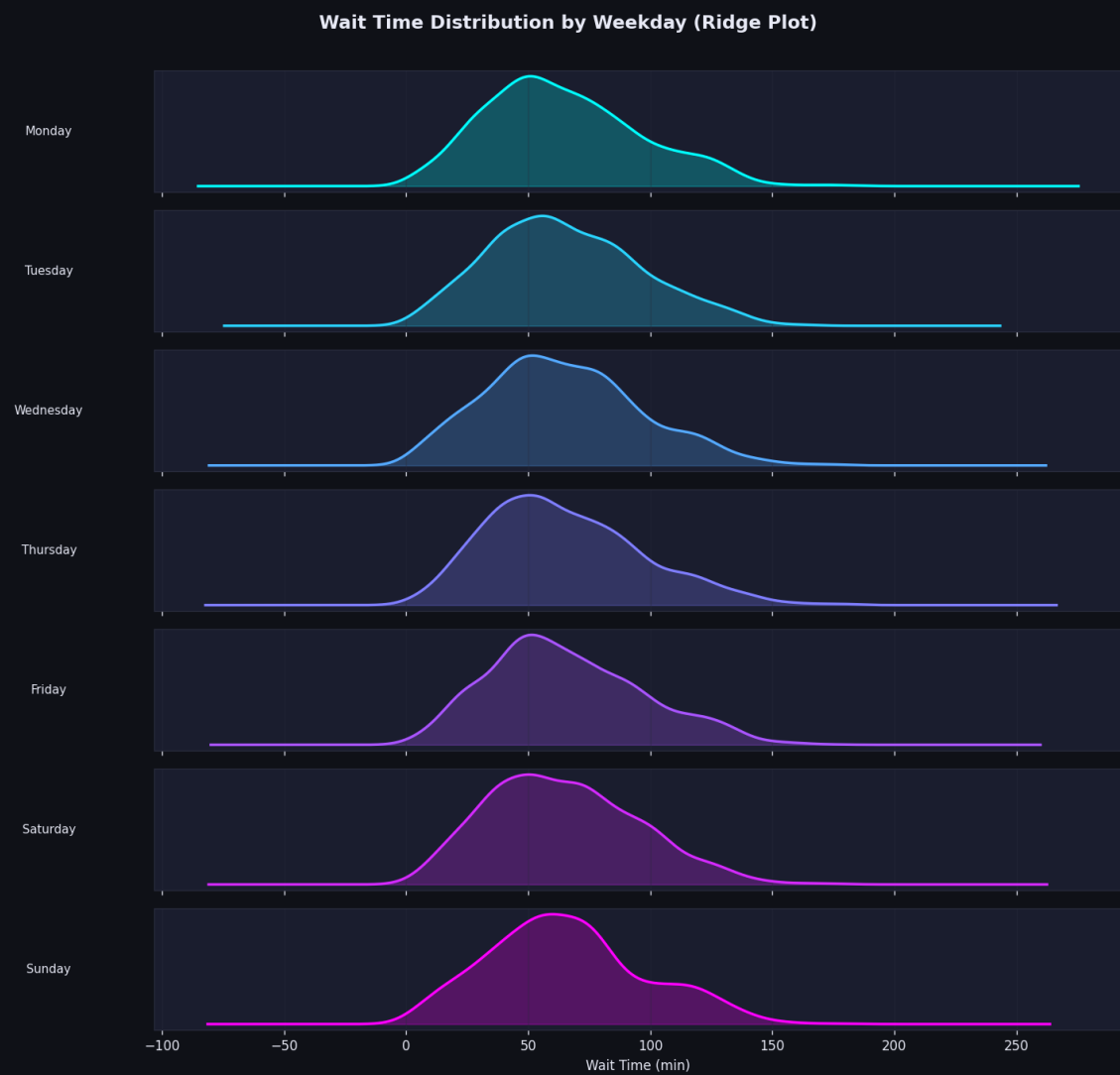
Wait Time Breakdown by Time Slot



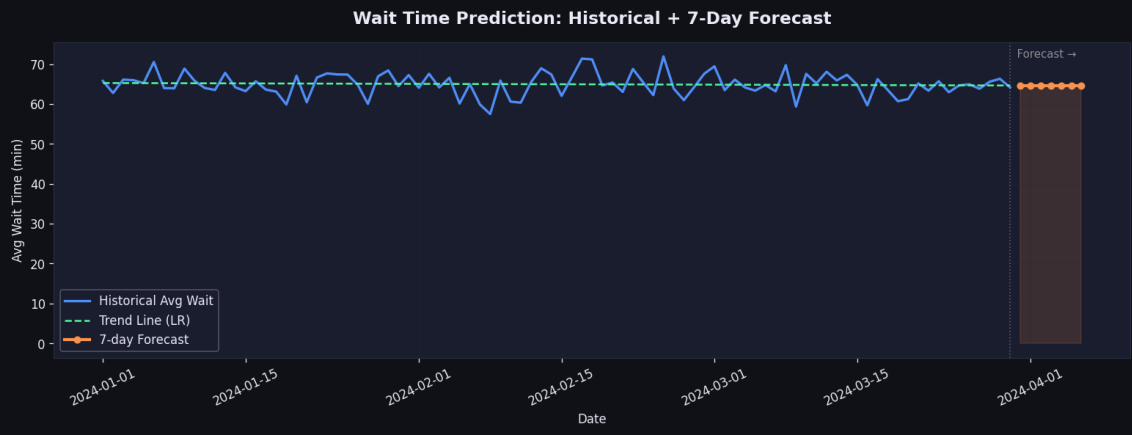
Daily Wait Time Trend



Weekday Wait Distribution (Ridge Plot)



7-Day Wait Time Prediction



## ■ Key Insights & Recommendations

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### ■ *Neurology has the highest average wait*

→ Average wait time is 102 min. Recommend adding 1–2 extra staff during peak hours.

### ■ *Neurology causes max cascading delay*

→ On stressed days, wait time jumps by ~18 min extra. Doctor cover plan needed.

### ■ *50 extreme delay events detected*

→ Extreme delays ( $>2.5\sigma$  above dept mean) suggest occasional system failures or data entry errors. Audit required.

### ■ *Neurology: peak at 8:00*

→ Avg wait reaches 129.0 min. Schedule extra consultants between 7:00–10:00.

### ■ *Cardiology: peak at 11:00*

→ Avg wait reaches 113.9 min. Schedule extra consultants between 10:00–13:00.

### ■ *Orthopedics: peak at 9:00*

→ Avg wait reaches 95.0 min. Schedule extra consultants between 8:00–11:00.

### ■ *Friday is the busiest day*

→ ~1599 patients on average. Consider full staffing + floating doctors on this day.

## ■ 7-Day Wait Time Forecast

| Date       | Day       | Predicted Wait (min) |
|------------|-----------|----------------------|
| 2024-03-31 | Sunday    | 64.6                 |
| 2024-04-01 | Monday    | 64.6                 |
| 2024-04-02 | Tuesday   | 64.6                 |
| 2024-04-03 | Wednesday | 64.6                 |
| 2024-04-04 | Thursday  | 64.6                 |
| 2024-04-05 | Friday    | 64.6                 |
| 2024-04-06 | Saturday  | 64.6                 |

Model Performance → MAE: 2.24 min | R²: 0.004 | Trend: ■ Stable

## ■ Doctor Absence Impact Analysis

Most impacted department: **Neurology** (+18.4 min cascading delay)

| Department  | Normal Avg | Stressed Avg | Cascading Delay | Stressed Days |
|-------------|------------|--------------|-----------------|---------------|
| Neurology   | 100.5 min  | 118.9 min    | +18.4 min       | 9             |
| Cardiology  | 85.6 min   | 99.4 min     | +13.8 min       | 9             |
| Orthopedics | 70.8 min   | 84.2 min     | +13.5 min       | 9             |
| Emergency   | 64.1 min   | 77.4 min     | +13.3 min       | 9             |
| Gynecology  | 57.9 min   | 69.4 min     | +11.5 min       | 9             |
| Pediatrics  | 49.7 min   | 61.3 min     | +11.5 min       | 9             |
| General Opd | 43.5 min   | 54.8 min     | +11.3 min       | 9             |
| Dermatology | 36.1 min   | 44.9 min     | +8.8 min        | 9             |